Measuring Finite-Budget Coordination-Collapse Crossings in Reward-Mixed Cooperative MARL*

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Abstract

Cooperative multi-agent reinforcement learning can fail when local incentives outweigh the team objective. We introduce a protocol for measuring where this failure occurs as the reward mixture changes. The protocol combines behavioral collapse labels, observed failure proportions, and seed-block uncertainty estimates to locate the 50% collapse crossing, w_{50}. On an inventory-network testbed with shared-parameter PPO, the crossing is 0.666 $[0.631, 0.696]$ at 300k agent transitions. Separating reward curvature from a loss-branch multiplier reveals opposing shifts: curvature lowers the crossing, while increasing the multiplier changes it by $+0.268$ $[+0.230, +0.306]$ relative to the curvature-only baseline. An independent longitudinal cohort retains a positive multiplier-minus-curvature difference through 3.6M transitions, while recovery analysis shows that collapse is not absorbing. A direct symmetric-scaling control moves the crossing at least as far as the loss-only multiplier, and the asymmetric-minus-symmetric interval includes zero. The experiment therefore identifies sensitivity to the transformed training signal, not a gain–loss-asymmetry mechanism. These results establish a comparative measurement tool for reward design; numerical crossings remain conditional on the task, learner, training budget, reward units, optimizer, and behavioral label.

1 Introduction

A team of learning agents can fail even when the task has a good cooperative solution: instead of learning complementary behavior, agents can settle into locally safe, low-cooperation behaviour. We call this *coordination collapse*; in the MARL literature its endpoint is the *lazy-agent* policy (Liu et al., 2023; Shi & Peng, 2023), and the underlying private-versus-collective conflict is the one studied in sequential social dilemmas (Leibo et al., 2017). We study this failure through a *reward-centralization* dial: within each reward-mixing sweep, agent i is trained on the blended scalar $x_i(w) = (1 - w)\, r_i^{\text{local}} + w\, r^{\text{global}}$, $w \in [0, 1]$, so at w=0 the learner sees only its own local reward and at w=1 only the team reward. Turn w down and an agent's private performance drowns out its contribution; cooperating earns no learnable signal and the lazy policy wins. That decentralized reward shaping cannot match centralized training is broadly accepted (Akella, 2025), and lazy-agent and credit-assignment work (Liu et al., 2023; Shi & Peng, 2023; Sunehag et al., 2018; Rashid et al., 2018; Foerster et al., 2018) mitigates the pathology; what none of it supplies is the *location*. The closest phase-diagram studies place their critical points on *environmental* (Yamaguchi, 2025) or opinion-dynamics (Capannoli et al., 2026) axes rather than reward structure, and none estimates a confidence-intervalled crossing in reward-centralization space for gradient-trained cooperative policies. Our central question is therefore twofold: does coordination collapse admit a *measurable* finite-budget behavioral crossing w_{50}, and how do reward transformations, including their scale, move that crossing? The first is the paper's primary measurement problem; the second is an application that also tests what the instrument can and cannot identify.

Pinning down this number matters because reward design is not neutral. w_{50} is not universal: it is a measured property of an environment, learner, horizon, and collapse label that tells a designer where their

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own system crosses into collapse. We estimate it on a six-node inventory network under shared-parameter PPO and re-measure it across horizons in an independent longitudinal cohort. We use prospect theory only as a controlled reward-transform family, not a psychological claim: precisely, a pointwise loss-asymmetric transform of the immediate blended reward, omitting the probability-weighting channel of full prospect theory. After blending, the scalar is transformed with value-function curvature and loss aversion (a loss looming larger than an equal gain (Kahneman & Tversky, 1979; Tversky & Kahneman, 1992)), while the learner, architecture, optimizer, and evaluation label are unchanged. A separate symmetric multiplier then tests whether a boundary shift is specific to gain–loss asymmetry or is also produced by scalar scaling.

We build the instrument before using it. Three design choices make a reading trustworthy: the collapse label is an explicit behavioral rule scored on evaluation; matched policies differ only in the scalar training signal, since the transform is applied after blending, outside the optimizer; and every w_{50} carries a bootstrap confidence interval. We use "boundary" in this operational sense: a finite-training, threshold-defined behavioral crossing in learning dynamics, not a claim of phase-like criticality.

The contributions are: (i) a protocol for locating a finite-budget behavioral crossing from observed collapse proportions, with empirical-bracketing checks and training-seed-block uncertainty; (ii) an independent checkpointed cohort that distinguishes cross-sectional boundary movement from within-policy recovery and retains a positive reward-transform contrast through 3.6M transitions; and (iii) a direct asymmetric-versus-symmetric multiplier experiment showing that the observed shift is not identified as gain–loss asymmetry. The contribution is not the logistic midpoint itself, which is standard response-curve machinery, but the controlled use of that estimand to compare MARL reward designs while exposing its dependence on budget, reward units, optimizer, and behavioral label.

2 Related Work

Cooperative MARL, studied since independent Q-learning on shared tasks (Tan, 1993; Buşoniu et al., 2008), inherits the lazy-agent and relative-overgeneralization failures in which agents abandon cooperation for a degenerate policy (Liu et al., 2023; Shi & Peng, 2023; Tampuu et al., 2017). The standard remedy is centralized credit assignment under CTDE, a centralized critic or factorized value function (Lowe et al., 2017; Yu et al., 2022; Sunehag et al., 2018; Rashid et al., 2018; Foerster et al., 2018; Son et al., 2019); related work ties the failure to decentralized exploration (Weng & Lee, 2026) or shows decentralized shaping cannot match centralized baselines (Akella, 2025). This establishes *that* centralization helps and *how* to implement it; none locates the threshold w_{50} or gives it a confidence interval, the gap we close, using w as the dial rather than a fixed factorization. Combined individual-and-team reward design has direct precedent: DE-MADDPG separates a per-agent and a team critic for exactly such mixtures (Sheikh & Bölöni, 2020), and packet-routing work has used RL routing as a testbed and later studied the lazy-versus-selfish trade-off with mixed and adaptive local/global signals (Boyan & Littman, 1994; Mao et al., 2020); but these set or adapt the mixture as a design choice rather than *locating* w_{50} or giving it a confidence interval.

IRAT learns individual- and team-reward policies with mutual discrepancy constraints (Wang et al., 2022). CLOT balances individual/team policy optimization through a dynamically updated Lagrangian multiplier under a policy-consistency constraint (Zhang et al., 2026). Our contribution is uncertainty-aware measurement of a finite-budget behavioral crossing, rather than reward balancing itself or a replacement policy-learning algorithm.

Statistically, a 50% crossing is an effective-dose or psychometric-threshold estimand rather than a new regression object. Dose–response analysis routinely estimates ED50 values and differences between them (Ritz et al., 2015), while psychometric work studies bootstrap uncertainty and the effect of grid placement on threshold localization (Wichmann & Hill, 2001). We import that logic into trained multi-agent populations. Our contribution lies in the experimental unit, collapse label, seed-block resampling, bracketing diagnostics, and paired comparison of reward designs.

Treating collective learning as a dynamical system with critical points has precedent: Yamaguchi (2025) builds a phase diagram for independent Q-learning over *environmental* axes, and Capannoli et al. (2026) prove a metastable boundary in opinion dynamics under a fixed update rule. Mean-field and dynamical-

Table 1: Primary experimental setup. The actor and critic use separate MLP heads but share one PPO policy instance, optimizer step, and global gradient-clip operation.

Item	Specification
Environment	Six-node `NetworkManagement-v1`; node 1 scripted, nodes 2–6 trained; 30-period episodes.
Local reward	Node i's period profit: sales revenue minus procurement, operating, holding, pipeline-holding, and unfulfilled-demand costs, with the environment's native discount.
Team reward	Sum of the five trained nodes' local period profits.
Reward mixture	$x_i(w) = (1-w)r_i^{\text{local}} + wr^{\text{global}}$; transform applied after mixing.
Policy architecture	One parameter-shared PPO policy across nodes; full joint observation; node-specific continuous action lanes. Actor and critic MLP heads are separate, but one Adam step and one global max-gradient-norm of .5 cover both.
Collapse label	$\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$, using held-out deterministic evaluation; $\bar{S}$ is mean base-stock action and $\bar{\pi}$ is ten-episode mean total profit over nodes 2–6.
Budgets	300k for the primary sweep and direct control; independent cohort checkpointed every 150k through 3.6M requested agent transitions.
Transform range	Linear null; curvature $\alpha = \beta = .88$ with $\lambda \in \{1, \ldots, 7\}$; direct control uses curvature-only, losses-only $\times 7$, and both-branches $\times 7$.
Uncertainty	Firth logistic midpoint with 2,000 percentile bootstrap draws of complete training-seed blocks; paired contrasts reuse each draw.

systems analyses derive deterministic or distributional limits for multi-agent learning (Hu et al., 2019; Barfuss et al., 2019; Leonte et al., 2025). We borrow the phase-diagram measurement style but differ in object: ours is a *reward-structure* threshold with confidence intervals, estimated from trained populations rather than asserted to be a dynamical critical point.

Human preference structure has entered collective learning only narrowly. Hughes et al. (2018) show inequity aversion improves cooperation within the cooperative-AI agenda (Dafoe et al., 2020; Hammond et al., 2025; Barfuss et al., 2020), and loss aversion is now a training objective at scale: KTO aligns language models with a Kahneman–Tversky value function carrying an explicit loss-aversion coefficient (Ethayarajh et al., 2024). Risk-sensitive RL optimizes tail outcomes (Qiu et al., 2021; Shen et al., 2023; Prashanth, 2014; Tamar et al., 2015) (distinct from loss aversion's reference-point asymmetry), and prospect theory has entered single-agent RL with known calibration concerns (Prashanth et al., 2016; Rabin, 2000; Tversky & Kahneman, 1992). In multi-agent settings, Mengesha et al. (2025) find loss aversion *non-monotone* in an evolutionary quantal-response model, and Lalmohammed et al. (2025) bring CPT into cooperative MARL but fold it into the actor–critic update, varying value and probability-weighting components jointly with the optimizer. We differ at the level of *apparatus*: rather than proposing another behavioral model, we build a measured boundary and use it as a common instrument; the prospect-theoretic reward-transform study is its first application, not the instrument's whole contribution.

3 Preliminaries and Method

We study *reward-mixed* cooperative MARL, in which each learner's scalar reward is the blend $x_i(w) = (1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}$, $w \in [0,1]$; our supply-chain learner is shared-parameter PPO (a shared actor–critic whose actor receives the full network observation and uses node-specific action indices). Here w is not an optimizer hyperparameter but a structural property of the reward design. We do not rescale r_i^{local} or r^{global} separately at each w; both stay in the environment's native units, so a given w means the same mixture in every condition rather than hiding a per-condition normalization. Unless stated otherwise, each experiment varies one factor at a time, holding the learner, environment, horizon, transform parameters, and label fixed.

The boundary estimate follows four steps: fix the environment, learner, horizon, and collapse label; train matched policies across a grid of w; label each run from pre-specified *evaluation* behavior, not training reward; and fit a monotone binary model $P(\text{collapse} \mid w)$, reporting the midpoint w_{50} with bootstrap confidence intervals.

The reward transform is prospect-theoretic in value-function form: with zero as the gain/loss reference, $v(x) = x^\alpha$ for $x \geq 0$ and $-\lambda(-x)^\beta$ for $x < 0$. The exponents $\alpha, \beta \in (0, 1]$ control *curvature* (values below one make the transform sublinear, compressing larger magnitudes); the separate multiplier $\lambda > 1$ scales the entire loss branch. For $a > 0$, $|v(-a)|/v(a) = \lambda a^{\beta-\alpha}$. This equals λ at every magnitude only when $\alpha = \beta$, as in our experiments. The case α=β=1, λ=1 leaves $v(x) = x$. Because curvature is present in every loss-aversion condition, we run a curvature-only baseline so its effect is not silently folded into the λ shift. We omit the probability-weighting channel of full CPT, so only the scalar value transform varies between matched runs. Applied after blending,

$$r_i^{\mathrm{PT}} = v\big((1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}\big)\,, \tag{1}$$

The controls implement $r_i^{\mathrm{PT}} = v((x_i - z_0)/s_{\mathrm{in}})/s_{\mathrm{out}}$, with positive input/output scales and no addition of z_0 after transformation. Equation 1 sets $z_0 = 0$ and both scales to one. Within each fixed configuration, this preserves the optimizer and architecture. This differs from Lalmohammed et al. (2025), who embed CPT *inside* the actor–critic update while varying the transform, probability weighting, and optimizer together. A measured shift is thus attributed to the transform channel, while recognizing that reward scale, variance, and advantage shape are part of that channel.

Training budgets count requested SB3 agent transitions; vectorization and rollout rounding determine actual joint environment ticks (Appendix A).

A *collapse label* is a pre-specified binary evaluation rule marking runs whose behavior lies inside a low-cooperation region, scored on held-out evaluation episodes rather than training reward. In the supply chain, collapse is $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$, with $\bar{S}$ the mean base-stock action at one deterministic seeded reset, averaged across the five trained nodes, and $\bar{\pi}$ their total episode profit averaged over ten deterministic evaluations; this corresponds to low/no ordering with poor profit. Both the team-reward sum and this profit metric cover nodes 2–6; node 1 is scripted to issue total quantity 20 across its action entries. The exact reward and observation conventions are in Appendix A. Collapsed runs pile up at the lazy-policy deterministic profit floor (≈ -128), so the profit clause is effectively a redundancy check on $\bar{S} < 10$ rather than a tuned cutoff. Threshold robustness over estimable $\pm 20\%$ variants changes w_{50} by at most 0.015.

We write the estimand as $w_{50}(T; E, A, V, U, L, Q)$, conditional on training budget T, environment E, learner A, transform V, reward units U, collapse label L, and evaluation protocol Q; w_{50} replaces the legacy notation w_{crit}. It is a fitted 50% crossing, not a safe operating point. Because w weights team reward, a rightward shift requires more team weight to reach the crossing.

For the primary supply sweep and horizon analysis, we use Firth logistic regression (Firth, 1993) consistently, fitting $P(\text{collapse} \mid w) = \sigma(b_0 + b_1 w)$ and reporting $w_{50} = -b_0/b_1$. Intervals use 2,000 bootstrap draws of training-seed blocks, retaining all sampled w-cells and using the same draw for paired contrasts (Efron & Tibshirani, 1993). They condition on the fixed collapse label. We export fit status, slopes, observed bracketing cells and every bootstrap result, without clipping out-of-grid estimates. Logistic, probit and decreasing isotonic sensitivity fits use the same input rows. The maximum estimator spread in the confirmatory supply-chain data is .018. An excluded exploratory transfer audit produced a larger spread of .158; this is why agreement is checked rather than assumed. Appendix B states the numerical checks and isotonic convention. The transition width is the fitted 10–90 width $\Delta w = (\mathrm{logit}(0.9) - \mathrm{logit}(0.1))/|b_1|$. The decomposition compares, at identical environment and optimizer, a **linear baseline** (α=β=1, λ=1), a **curvature baseline** (α=β=0.88, λ=1), and **loss-aversion conditions** (α=β=0.88, $\lambda > 1$). The primary substrate is `NetworkManagement-v1` (Hubbs et al., 2020), a six-node multi-echelon inventory network (Oroojlooyjadid et al., 2022; Gijsbrechts et al., 2022; Leluc et al., 2023), chosen because it exposes a directed-flow coordination failure. Agents run shared-parameter PPO (Schulman et al., 2017) via Stable-Baselines3 (Raffin et al., 2021), wrapped with PettingZoo/SuperSuit (Terry et al., 2021; 2020). Headline λ sweeps use 20 seeds per (w, λ) cell; Appendix B reports the observed cells.

4 Observed Crossings and Estimator Validation

A well-defined boundary exists at fixed training budget. At 300k steps on the supply-chain PPO substrate, collapse has a clear, tightly-estimated boundary in reward-centralization space (Figure 1a): the

recomputed midpoint is 0.666 $[0.631, 0.696]$. The displayed observed proportions and Wilson intervals make the sampled crossing support visible. Logistic, probit and isotonic sensitivity results are reported on the same rows; agreement is not assumed uniform across settings.

"Collapse" is an operational threshold crossing, not a confirmed attractor. A fitted midpoint requires an identified nonzero slope; empirical bracketing must be checked separately. We tested the underlying continuous behavior for a distinct dynamical regime: bootstrap 95% CIs on Sarle's bimodality coefficient (Sarle, 1983) and Hartigan's dip test (Hartigan & Hartigan, 1985) against a unimodal null. On the supply chain Sarle's coefficient rises near the crossing (0.59 at w=0.65 and 0.57 at w=0.75, both above the 0.555 rule-of-thumb) but its CI straddles the threshold everywhere, and the dip test never rejects unimodality ($p \geq 0.30$). The tested data therefore do not clear a formal multimodality test, so we use *threshold-defined behavioral crossing* throughout.

5 Training-Budget Dependence

The boundary is finite-horizon. The boundary slides left across four linear-reward budgets: the recomputed midpoints are 0.821, 0.666, 0.622 and 0.572 at 150/300/450/900k requested agent transitions (Figure 1). These are separate finite-budget estimates; neither a nonzero asymptote nor within-policy recovery follows from this cross-sectional comparison.

Independent longitudinal confirmation. After 120 pilot trajectories, we froze five common cells $w \in \{.2, .4, .6, .85, 1\}$ and ten disjoint confirmation seeds per cell for linear, curvature-only and $\lambda = 7$ rewards: 150 trajectories to 3.6M, evaluated every 150k. The loss-multiplier-minus-curvature crossing difference remains positive at all four primary budgets: +.281 $[.174, .363]$, +.284 $[.182, .393]$, +.326 $[.222, .433]$, and +.146 $[.072, .229]$ at 300k, 900k, 1.8M, and 3.6M (Figure 2a). These pointwise seed-block intervals support persistence over the tested budgets, not an infinite-horizon limit. None of the three conditions meets the declared practical-stability criterion: the entire interval for its 3.6M-minus-1.8M change must lie within $[-.02, .02]$. At 1.8M, the absolute crossings are .589 $[.500, .698]$ (linear), .399 $[.291, .503]$ (curvature), and .725 $[.725, .725]$ (loss multiplier); at 3.6M they are .590 $[.482, .700]$, .524 $[.420, .630]$, and .670 $[.604, .725]$. The crossing has not demonstrably stopped moving. The zero-width 1.8M multiplier interval reflects identical seed-label patterns on a coarse grid, not exact localization.

Recovery and evaluation sensitivity. Checkpoints track the same policies, permitting recovery and relapse analysis conditional on the 300k label. Across the fixed grid, 6/27 linear, 4/23 curvature-only and 7/35 lambda-7 trajectories collapsed at 300k are noncollapsed at 3.6M; among initially noncollapsed trajectories, 3/23, 2/27 and 0/15 respectively relapse. These descriptive totals are stratified by mixture cell in Appendix C, alongside three-checkpoint recovery and subsequent relapse. Increasing evaluation from ten to fifty episodes at 900k, 1.8M and 3.6M changes only one of 450 labels and leaves the 3.6M contrast unchanged. Coarsening has a cost: identical seed-label patterns give zero-width bootstrap intervals at several $\lambda = 7$ checkpoints, including 900k and 1.8M. Their observed crossing bracket is still $[.6, .85]$; bootstrap degeneracy does not imply exact localization. Appendix C gives full checkpoint curves, cell support and sensitivity tables.

6 Reward-Transform Comparisons

The selected loss-multiplier shift opposes curvature. Two baselines anchor the comparison, all with identical environment, learner, label, and 300k horizon. The linear null (α=β=1, λ=1) was measured twice as a reproducibility check (0.666 $[0.631, 0.696]$ in the main structural sweep and 0.685 $[0.650, 0.715]$ in the independent decomposition run, overlap does not establish equivalence; the decomposition uses its own linear run); the second is the curvature baseline (α=β=0.88, λ=1), fixed at the conventional prospect-theory curvature estimate (Tversky & Kahneman, 1992), not tuned; the sweep then varies only $\lambda \in \{1, \ldots, 7\}$. The selected high-minus-low endpoint is positive, with high-lambda estimates near 0.85 within the sampled grid (Figure 3, Table 2). Curvature alone acts in the *opposite* direction, lowering w_{50} to 0.585 (−0.100

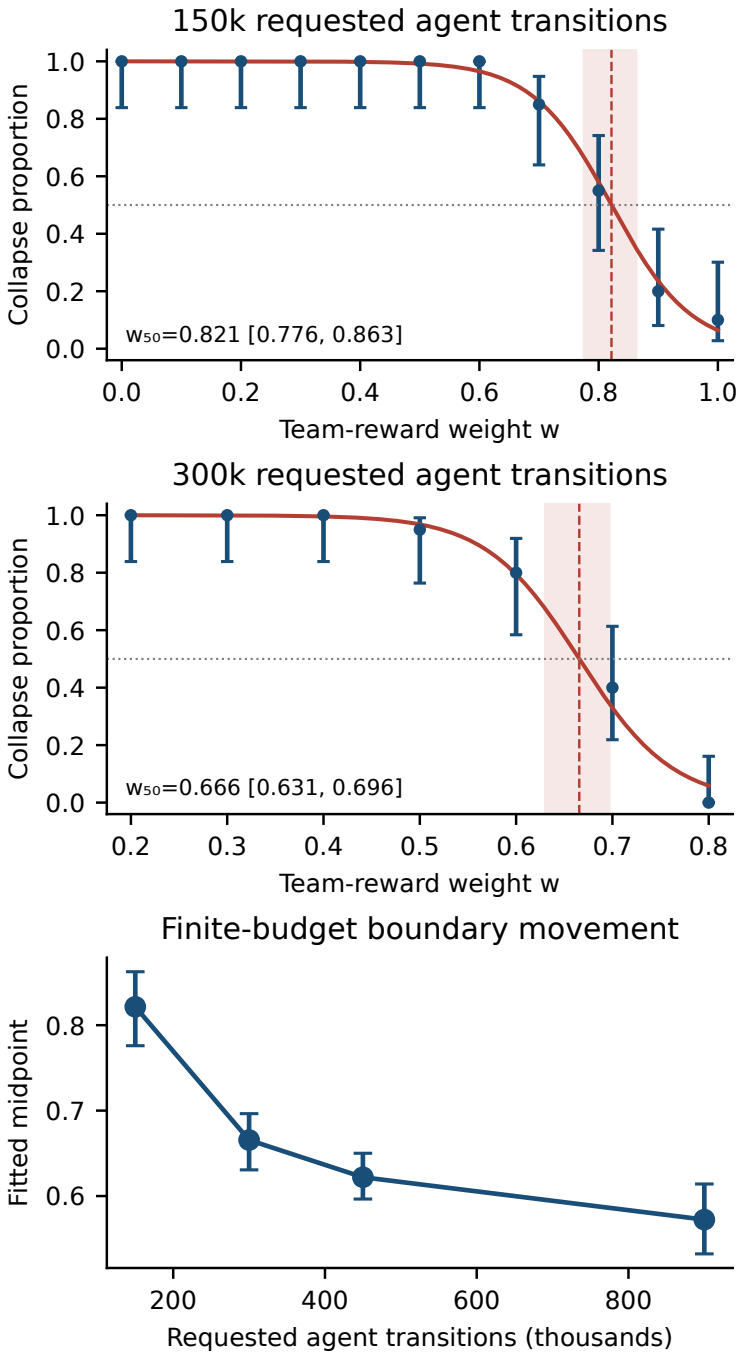


Figure 1: Finite-budget supply PPO crossings. Top and middle: observed collapse proportions with 95% Wilson intervals, Firth curves drawn only over sampled grids, midpoint lines and seed-block 95% CI bands. Bottom: independently fitted horizons; budgets count requested agent transitions.

$[-0.145, -0.057]$ vs. linear), which is why we decompose: as a single "prospect-theoretic" treatment the two effects partially cancel, since here curvature lowers the measured crossing while the loss multiplier raises it.

Two effect sizes answer two questions. The effect size depends on the baseline (Table 2). The *full-transform shift* against the linear baseline is the total change a designer sees adopting the full transform;

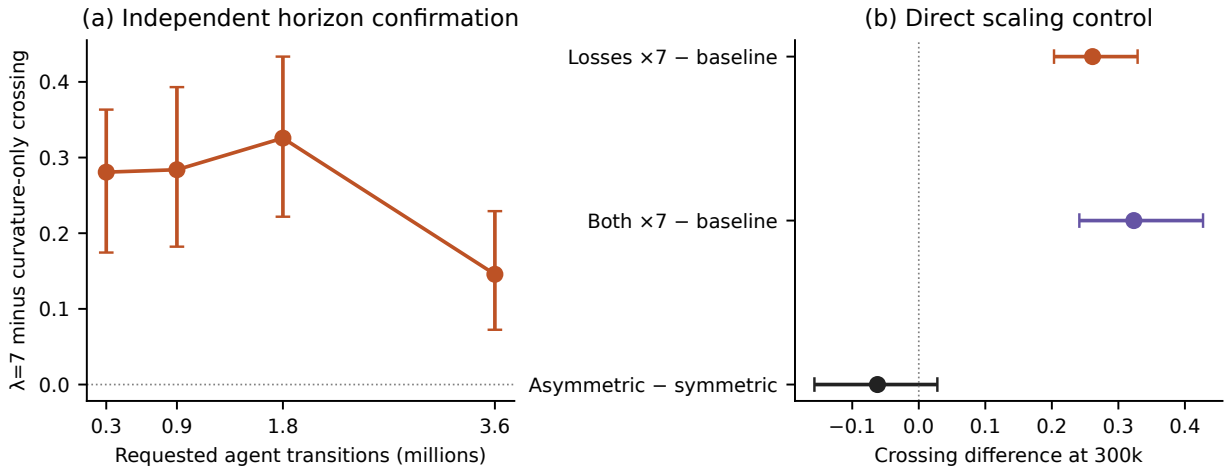


Figure 2: Independent confirmation with ten paired training seeds and 2,000 seed-block bootstrap draws. (a) Loss-multiplier-minus-curvature crossing differences at four declared budgets, using five common mixture cells and reference zero. (b) Three-arm control at 300k on eight cells, with a shared empirical-mean reference: both asymmetric and symmetric multipliers shift the crossing relative to curvature-only, while their direct difference includes zero. Intervals are pointwise; the two panels use different fixed references and are not pooled.

the *loss-multiplier shift* against the curvature baseline is the cleaner estimate of the λ effect, since curvature is present in every sweep condition, and the two differ by the curvature offset (-0.100). Throughout, the loss-multiplier effect denotes this increment; it does not by itself identify gain–loss asymmetry.

Reference and scale sensitivity. The loss multiplier changes both sign weighting and reward scale. A fixed pilot-RMS control leaves a recomputed endpoint difference of $+.145$ $[.108, .186]$, but fixed calibration does not ensure achieved on-policy RMS or advantage matching. The common reward-unit coordinate $q = wc_g/[(1-w)c_\ell + wc_g]$, where c_ℓ, c_g are applied unit multipliers, also retains a local-unit difference of $+.053$ $[.009, .100]$ at lambda 1. We therefore completed a direct three-arm comparison on eight mixture cells with ten seeds each, holding $z_0 = -2.6281398333$ fixed from 60,000 independent calibration samples. With $u = x - z_0$ and $\phi(u) = \text{sign}(u)|u|^{.88}$, the arms are $B = \phi(u)$, $A = v_7(u)$ and $S = 7\phi(u)$. Their crossings are .560, .821 and .883. Relative to B, A shifts the crossing by $+.261$ $[.203, .329]$ and S by $+.323$ $[.241, .427]$; the direct A–S difference is $-.062$ $[-.157, .028]$ (Figure 2b). This does not establish an asymmetry-specific effect or equivalence between arms.

Both gain and loss branches are active: near-crossing loss fractions are 34.0%, 31.1% and 20.4% for B, A and S. Nevertheless, pooled transformed-reward RMS values are 8.80, 30.36 and 57.67, and raw advantage SDs are 35.05, 135.95 and 300.40. Normalized optimizer advantages have almost identical SD (.99765), but every control minibatch triggers global gradient clipping, with critic gradients dominating. This leaves an optimizer-scaling pathway despite advantage normalization; it is not a causal diagnosis. Appendix D gives observed cells, intervals and training diagnostics.

Equivalent linear-training budget. As a descriptive effect-size translation, we fit the four cross-sectional linear estimates to $w_{50} = 1.4824 - 0.13748\log(T/1000)$ and invert that curve for each transform crossing. Curvature-only at 300k maps to a linear budget of 685k $[535, 942]$, whereas $\lambda = 7$ maps to 98k $[67, 128]$; the repeated linear null maps to 330k $[275, 406]$. Thus the observed $\lambda = 7$ crossing resembles roughly one third of the nominal linear-training budget under this local empirical mapping. The $\lambda = 7$ value extrapolates below the observed 150k–900k horizon range, and none of these mappings identifies a causal equivalence.

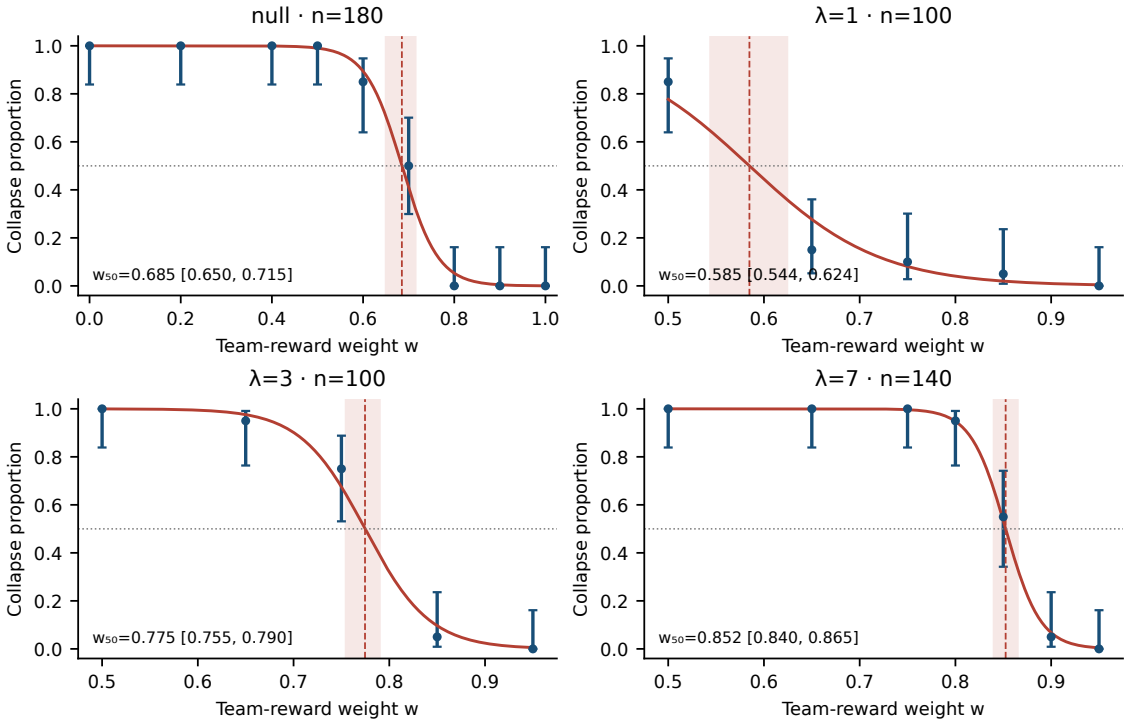


Figure 3: Supply PPO null and selected transform conditions at 300k. Points show collapse counts divided by cell totals with 95% Wilson intervals; curves are Firth fits over observed grids. Vertical bands are midpoint seed-block intervals. The lambda-7 panel includes the completed $w = .8, .9$ refinements; Appendix B reports the full sweep counts.

Table 2: Supply-PPO reward-transform decomposition at 300k requested agent transitions and descriptive equivalent linear-training budgets. Differences use paired seed-block draws. Equivalent budgets invert the four-point cross-sectional fit $w_{50} = 1.4824 - 0.13748 \log(T/1000)$ and propagate the same 2,000 bootstrap replicates; they are descriptive, not causal. † denotes extrapolation below the observed 150k–900k horizon range.

Condition	Crossing and 95% CI	Versus null	Versus curvature	Equivalent budget, k
Linear null	$0.685\ [0.650, 0.715]$	–	–	$330\ [275, 406]$
Curvature ($\lambda = 1$)	$0.585\ [0.544, 0.624]$	$-0.100\ [-0.145, -0.057]$	–	$685\ [535, 942]$
$\lambda = 3$	$0.775\ [0.755, 0.790]$	$+0.090\ [+0.065, +0.115]$	$+0.190\ [+0.150, +0.229]$	$172\ [140, 211]$
$\lambda = 7$	$0.852\ [0.840, 0.865]$	$+0.167\ [+0.134, +0.203]$	$+0.268\ [+0.230, +0.306]$	$98\ [67, 128]^{\dagger}$

The curvature result is scoped to supply-chain PPO. A loss-multiplier effect is a property of the transformed training signal; the completed direct control does not identify a sign-dependent effect beyond scalar scaling. Neither an endpoint difference nor the hypothetical SNR account supplies that identification on its own.

Cost and normalization. A five-cell, 20-seed boundary requires 100 trained policies. Retrospective fixed-grid thinning saves 20–40% of runs in the tested 20-seed comparisons, but can change the midpoint by .074; it does not validate an adaptive design or guarantee accuracy in a new domain. Confirmation consumed 530.57 CPU-hours for the horizon cohort and 61.38 for direct controls, including evaluation and checkpointing. Appendix E reports elapsed time, pilot costs and memory. Fix local/team units, sum-versus-average aggregation and reference transformation before comparison; the common q coordinate does not remove optimizer-scale sensitivity.

7 Discussion

Separating curvature from the loss multiplier gives opposing crossing shifts on supply-chain PPO, but does not separate asymmetry from optimizer-scale effects. The independent long-horizon comparison retains a positive multiplier contrast through 3.6M; the empirical-mean direct control also finds a positive shift under symmetric scaling. Together these results support the measurement instrument and the budget-conditional transform comparison, while leaving an asymmetry-specific mechanism unestablished.

The crossing is a diagnostic of the specified training and evaluation procedure. A deployment requirement such as at most 5% collapse would need its own probability target and uncertainty assessment; the 50% midpoint supplies no such guarantee. The collapse definition and its observed behavioral components are reported in Appendix A.

Scope and Future Work. Three limits point to next steps. The quantitative core is single-substrate, so magnitudes belong to one simulator and must be re-estimated elsewhere. The direct control does not isolate asymmetry from scale, and critic-dominated global clipping supplies a concrete but untested causal pathway. A targeted optimizer intervention is therefore more informative than another descriptive transform sweep. Which implementation choices make collapse persistent is not identified by these data; resolving that question requires common-code ablations rather than comparison with a separately rebuilt pipeline. Our crossing therefore conditions explicitly on the policy architecture, joint clipping, rollout rounding, observation wrapper, and simulator stream recorded in Appendix A. Localizing w_{50} from dynamics also resisted two probes, so a predictive observable remains an open problem. The common reward-unit coordinate retains measured differences; fixed units and aggregation conventions remain part of the protocol. Retrospective grid thinning saves runs but can move the midpoint appreciably (up to .074 for the tested 20-seed primary thinning rule); this is not evidence of universally cheap boundary estimation.

8 Conclusion

Reward centralization is usually treated as a direction: more team signal helps. We show that it also has a measurable finite-budget location, w_{50}, and that the estimated crossing changes with reward transforms and training budget. Independent confirmation retains a multiplier-minus-curvature shift through 3.6M transitions, but a symmetric-scaling control moves the crossing at least as far as the loss-only multiplier. The evidence therefore supports a comparative measurement protocol and a reward-scale sensitivity result, not a gain–loss-asymmetry mechanism. The numerical threshold remains conditional on the environment, learner, optimizer, budget, reward units, and behavioral label.

Ethical Statement

This work is entirely simulated, with no human subjects or personal data. Reported thresholds are substrate-specific and must be re-measured before use.

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A Experimental Configuration

Environment and learner. The primary environment is the six-node `NetworkManagement-v1` inventory network. Node 1 is scripted; nodes 2–6 are controlled by one parameter-shared PPO policy. The padded joint observation is visible to the actor, while each vector lane supplies the action for one trainable node. Unless a specific experiment states otherwise, PPO uses the Stable-Baselines3 defaults with `n_steps`= 128, minibatch size 256, ten optimization epochs, learning rate 3×10^{-4}, discount .99, GAE coefficient .95, policy clip .2, value-loss coefficient .5, zero entropy coefficient, normalized minibatch advantages, and global gradient max-norm .5. Training is CPU-only and uses a single explicit simulator random stream per environment.

Reward and label. Each agent receives $x_i(w) = (1-w)r_i^{\text{local}} + wr^{\text{global}}$ before any transform. The supply-chain collapse label is $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$, where $\bar{S}$ is the mean deterministic base-stock action across the five trained nodes and $\bar{\pi}$ is their ten-episode mean total profit. The label is evaluated independently of the transformed training reward. Its profit clause is a redundancy check because collapsed policies concentrate at the deterministic lazy-policy floor near -128. Evaluation seed blocks are $10s, \ldots, 10s+9$ for training seed s.

Transform conditions. The linear condition uses $v(x) = x$. Curvature-only uses $\phi(x) = \text{sign}(x)|x|^{.88}$. The loss-multiplier condition uses $v_\lambda(x) = \phi(x)$ for $x \geq 0$ and $\lambda\phi(x)$ for $x < 0$. The independent direct control uses the fixed reference $z_0 = -2.6281398333$, estimated before confirmation from 60,000 equally weighted blend samples produced by 40 untreated calibration policies. With $u = x - z_0$, its arms are $B = \phi(u)$, $A = v_7(u)$, and $S = 7\phi(u)$.

B Boundary Estimation and Diagnostics

For each fixed configuration we fit $P(Y = 1 \mid w) = \sigma(b_0 + b_1 w)$ by Firth logistic regression and report $w_{50} = -b_0/b_1$. A point estimate is reported only with a nonzero identified slope; observed probabilities are inspected separately for adjacent cells bracketing .5. Confidence intervals use 2,000 percentile-bootstrap draws of complete training-seed blocks. The same multiplicity vector is used across conditions in paired contrasts. Fits outside $[0, 1]$ and nonconverged or unidentified draws are retained and counted rather than clipped or silently replaced.

Logistic, probit, and decreasing isotonic fits are sensitivity estimators. The primary supply-PPO point-estimator spread is at most .018. The independent confirmation was also reproduced with a separate numerical optimizer to within 4×10^{-9} at all fifteen primary budget/condition points. Confidence intervals are pointwise; no simultaneous statement across checkpoints is made.

Table 3: Observed supply PPO collapse counts at 300k. Entries are $w : k/n$; Wilson intervals and all-setting cell records are exported in collapse_cells.csv.

Condition	Observed cells
null	0.00: 20/20, 0.20: 20/20, 0.40: 20/20, 0.50: 20/20, 0.60: 17/20, 0.70: 10/20, 0.80: 0/20, 0.90: 0/20, 1.00: 0/20
λ =1	0.50: 17/20, 0.65: 3/20, 0.75: 2/20, 0.85: 1/20, 0.95: 0/20
λ =2	0.50: 20/20, 0.65: 17/20, 0.75: 6/20, 0.85: 0/20, 0.95: 0/20
λ =3	0.50: 20/20, 0.65: 19/20, 0.75: 15/20, 0.85: 1/20, 0.95: 0/20
λ =4	0.50: 20/20, 0.65: 20/20, 0.75: 18/20, 0.85: 3/20, 0.95: 0/20
λ =5	0.50: 20/20, 0.65: 20/20, 0.75: 20/20, 0.80: 17/20, 0.85: 3/20, 0.90: 0/20, 0.95: 0/20
λ =6	0.50: 20/20, 0.65: 20/20, 0.75: 17/20, 0.80: 17/20, 0.85: 10/20, 0.90: 0/20, 0.95: 0/20
λ =7	0.50: 20/20, 0.65: 20/20, 0.75: 20/20, 0.80: 19/20, 0.85: 11/20, 0.90: 1/20, 0.95: 0/20

Table 4: Full supply PPO sweep after inclusion of all 120 completed refinement runs. Slopes are Firth estimates; no legacy Wald significance claims are retained.

λ	Runs	Midpoint and 95% CI	Slope	Usable
1	100	0.585 [0.544, 0.624]	−14.72	2000/2000
2	100	0.714 [0.683, 0.745]	−27.24	2000/2000
3	100	0.775 [0.755, 0.790]	−29.68	2000/2000
4	100	0.805 [0.780, 0.829]	−36.84	2000/2000
5	140	0.825 [0.812, 0.838]	−67.01	2000/2000
6	140	0.835 [0.816, 0.853]	−33.55	2000/2000
7	140	0.852 [0.840, 0.865]	−55.23	2000/2000

C Independent Longitudinal Confirmation

Pilot seeds 100–104 selected a common five-cell grid. The subsequent confirmation was frozen before execution and uses disjoint seeds 42–51, three reward conditions, and $w \in \{.2, .4, .6, .85, 1\}$. All 150 trajectories start from initialization, run to 3.6M requested agent transitions, and are evaluated every 150k transitions. Pilot outcomes do not enter the confirmation estimates. Every scheduled evaluation, configuration, transition counter, evaluation seed, source hash, and collapse label was checked before analysis.

C.1 Recovery, relapse, and evaluation sensitivity

Recovery and relapse are descriptive within-policy outcomes conditional on the 300k label. Sustained recovery is the first occurrence of three consecutive noncollapsed checkpoints after 300k and may be followed by relapse. These proportions describe the five fixed mixture cells rather than a task-level population distribution.

The fifty-episode sensitivity evaluation shares its first ten episodes with the primary evaluation and preserves training random state. Only one of 450 trajectory/checkpoint labels changes; the 1.8M and 3.6M contrasts are unchanged.

Table 5: Independent confirmation: ten seeds per condition and cell; 2,000 paired seed-block bootstrap draws. † marks identical seed-label patterns and zero bootstrap width; the empirical bracket remains [.6, .85]. Intervals are pointwise.

Budget	Linear	Curvature	Loss multiplier 7	Multiplier minus curvature
300k	0.648 [0.582, 0.725]	0.562 [0.500, 0.626]	0.843 [0.778, 0.893]	0.281 [0.174, 0.363]
900k	0.543 [0.500, 0.604]	0.441 [0.332, 0.543]	0.725 [0.725, 0.725]†	0.284 [0.182, 0.393]
1800k	0.589 [0.500, 0.698]	0.399 [0.291, 0.503]	0.725 [0.725, 0.725]†	0.326 [0.222, 0.433]
3600k	0.590 [0.482, 0.700]	0.524 [0.420, 0.630]	0.670 [0.604, 0.725]	0.146 [0.072, 0.229]

Table 6: Observed collapsed/total counts for the independent horizon confirmation at the four primary budgets. The same ten seeds occur at every cell and budget. All 24 checkpoint panels remain in the raw exports.

Budget	Condition	$w = .2$	.4	.6	.85	1
300k	Linear	10/10	10/10	7/10	0/10	0/10
300k	Curvature only	10/10	10/10	3/10	0/10	0/10
300k	Loss multiplier 7	10/10	10/10	10/10	5/10	0/10
900k	Linear	10/10	10/10	2/10	0/10	0/10
900k	Curvature only	9/10	7/10	1/10	0/10	0/10
900k	Loss multiplier 7	10/10	10/10	10/10	0/10	0/10
1800k	Linear	10/10	10/10	2/10	1/10	1/10
1800k	Curvature only	9/10	5/10	1/10	0/10	0/10
1800k	Loss multiplier 7	10/10	10/10	10/10	0/10	0/10
3600k	Linear	10/10	9/10	2/10	3/10	0/10
3600k	Curvature only	10/10	7/10	4/10	0/10	0/10
3600k	Loss multiplier 7	10/10	10/10	8/10	0/10	0/10

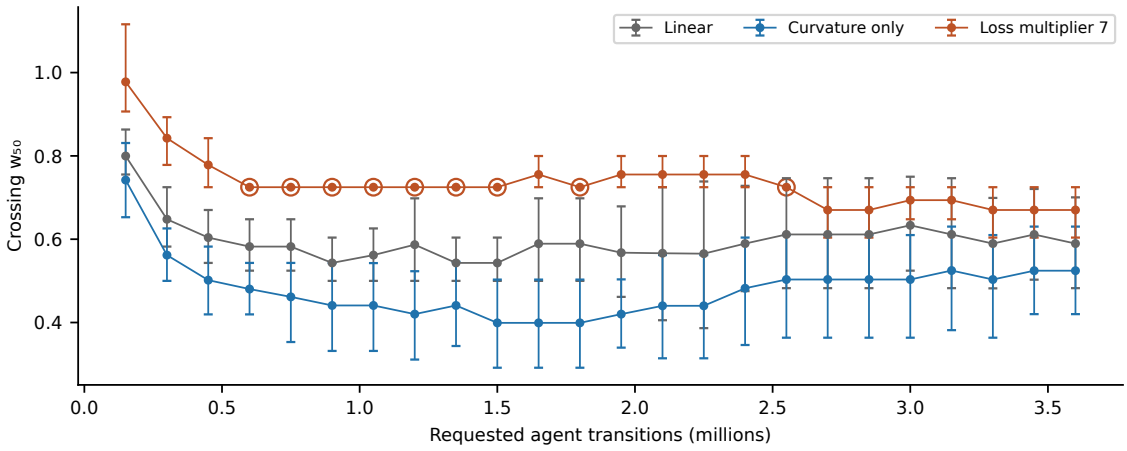


Figure 4: All independent-confirmation checkpoints with untruncated pointwise seed-block intervals. Open rings mark zero-width bootstrap intervals caused by identical seed-label patterns on the coarse grid; they do not imply exact localization.

Table 7: Within-policy outcomes conditional on the 300k label, by mixture cell. Recovery/relapse columns describe the 3.6M state. Sustained recovery means ever observing three consecutive noncollapsed checkpoints after 300k; it need not persist to 3.6M. A dash denotes a zero denominator.

Condition	w	Collapsed at 300k	Recovered at 3.6M	Ever sustained	Relapsed at 3.6M
Linear	0.2	10	0/10	1/10	–
Linear	0.4	10	1/10	1/10	–
Linear	0.6	7	5/7	5/7	0/3
Linear	0.85	0	–	–	3/10
Linear	1	0	–	–	0/10
Curvature only	0.2	10	0/10	1/10	–
Curvature only	0.4	10	3/10	5/10	–
Curvature only	0.6	3	1/3	2/3	2/7
Curvature only	0.85	0	–	–	0/10
Curvature only	1	0	–	–	0/10
Loss multiplier 7	0.2	10	0/10	0/10	–
Loss multiplier 7	0.4	10	0/10	0/10	–
Loss multiplier 7	0.6	10	2/10	2/10	–
Loss multiplier 7	0.85	5	5/5	5/5	0/5
Loss multiplier 7	1	0	–	–	0/10

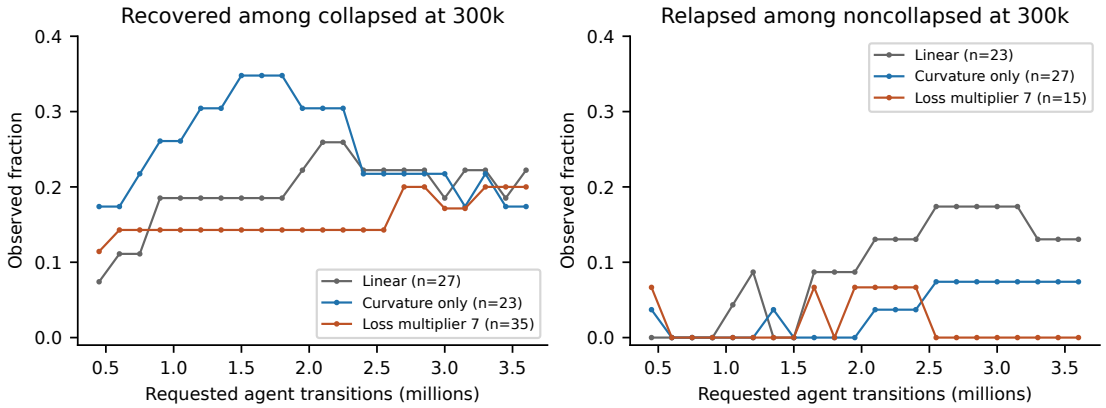


Figure 5: Observed recovery and relapse fractions conditional on the 300k label. Curves show current checkpoint state and do not establish an absorbing transition or convergence.

Table 8: Loss-multiplier-minus-curvature contrast under ten versus fifty evaluation episodes. Only one of 450 trajectory/checkpoint labels changes (curvature, $w = .4$, seed 51, 900k). The fifty episodes include the original ten and are a sensitivity check, not an independent replication.

Budget	Ten episodes	Fifty episodes
900k	0.284 [0.182, 0.393]	0.305 [0.204, 0.414]
1800k	0.326 [0.222, 0.433]	0.326 [0.222, 0.433]
3600k	0.146 [0.072, 0.229]	0.146 [0.072, 0.229]

D Direct Asymmetric-versus-Symmetric Control

The control cohort contains $3 \times 8 \times 10 = 240$ independently trained policies at 300k requested transitions. All arms use the same empirical-mean reference, mixture cells, training seeds, evaluation seeds, learner, and optimizer. Four of 2,000 symmetric-arm bootstrap crossings fall outside the sampled grid and remain in the interval calculation.

The paired differences are $A - B = +.261$ $[.203, .329]$, $S - B = +.323$ $[.241, .427]$, and $A - S = -.062$ $[-.157, .028]$. Thus both multipliers shift the crossing, whereas their direct difference establishes neither an asymmetry-specific effect nor practical equivalence.

Near the empirical crossing brackets, loss fractions are 33.966%, 31.096%, and 20.393% for B, A, and S. Across all cells, transformed-reward RMS values are 8.796, 30.359, and 57.673; raw rollout advantage standard deviations are 35.045, 135.948, and 300.400. Normalized optimizer-advantage standard deviation is .997653 in every arm. Nevertheless, every control minibatch triggers global gradient clipping. Mean pre-clip critic norms are 1,305, 3,622, and 12,267, compared with actor norms 2.04, 1.73, and 1.50. These diagnostics identify a critic-dominated scaling pathway but are not a causal intervention.

Table 9: Direct control at 300k with common empirical-mean reference, eight mixture cells and ten seeds; Firth fits with 2,000 seed-block draws.

Arm	Transform	Crossing and 95% interval
B	$\phi(u)$	0.560 [0.516, 0.600]
A	$v_7(u)$	0.821 [0.775, 0.864]
S	$7\phi(u)$	0.883 [0.821, 0.954]

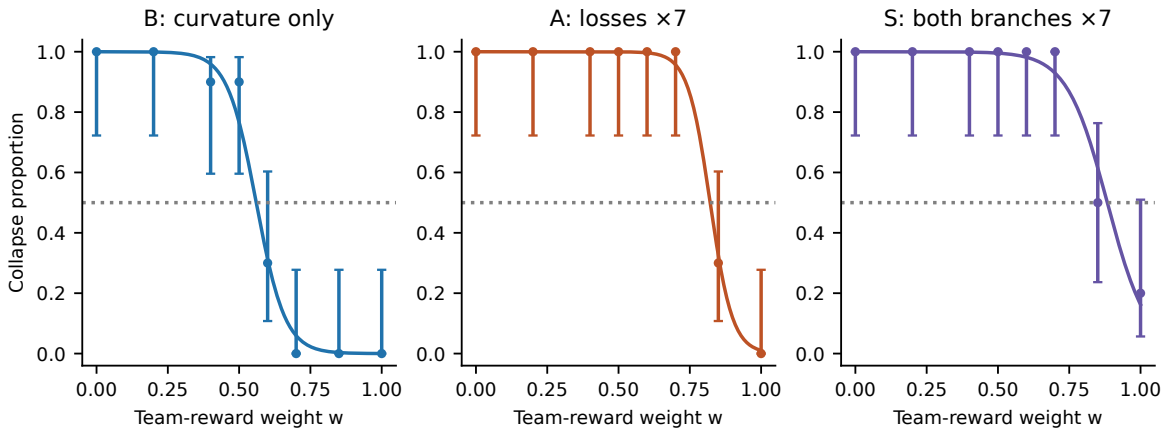


Figure 6: Observed collapse proportions for the three direct-control arms. Each point represents ten training seeds; bars are 95% Wilson intervals and curves are Firth fits over the sampled grid.

E Computational Accounting

Scheduler accounting sums completed worker CPU consumption and excludes array-parent records. Elapsed span is the interval from the first worker start to the last worker finish; concurrent spans are not added. Memory is the largest scheduler-reported per-worker peak, not cluster-wide peak memory.

The confirmation cohorts consumed 591.95 CPU-hours. Including corrected pilots, calibration, source gates, smoke tests, and diagnostic auditing, the captured new work consumed 1,005.89 CPU-hours. A superseded pilot consumed another 39.59 CPU-hours and is excluded from scientific cohorts. These are measured revision costs, not lifetime project costs.

Table 10: Measured final PBS resources for the new cohorts. Counts in parentheses are trajectories or calibration policies, not workers. CPU-hours sum worker consumption; elapsed span is wall-clock batch duration. Memory is the largest scheduler-reported worker peak. Initial and continuation stages refer to the same pilot trajectories.

Work	Workers	CPU-hours	Elapsed hours	Peak MiB/worker
Horizon confirmation (150)	32	530.57	27.78	547.0
Direct controls (240)	16	61.38	5.42	543.7
Pilot initial stage (105)	8	79.31	12.48	546.6
Pilot continuation (same 105)	32	275.35	16.24	1069.1
Pilot refinement (15)	15	57.93	5.21	547.0
Base calibration (35)	4	0.71	0.23	1069.9
Refined-cell calibration (5)	5	0.11	0.05	541.2
Full diagnostic audit	32	0.49	0.15	13.6

F Reproducibility Artifact

The anonymized supplementary archive contains the exact manuscript source, immutable executed-source snapshots and hashes for the new confirmation cohorts, their frozen designs and raw outputs, calibration records, analysis scripts, bootstrap inputs, generated tables and figures, and rebuild commands. The older primary sweep is accompanied by its audited per-run records, but no claim is made that its original execution environment was preserved. Exploratory settings that cannot be tied to an immutable executed configuration are excluded from the paper's confirmatory claims.